

12 (**short**) The linear difference equation

$$f_i = f_{i-1} + f_{i-2}, \text{ with } f_0 = 0 \text{ and } f_1 = 1$$

generates the Fibonacci sequence 0, 1, 1, 2, 3, 5, 8... The equation has solution

$$f_i = \frac{(\lambda_1)^i - (\lambda_2)^i}{\sqrt{5}}, \text{ where } \lambda_1 = \frac{1 + \sqrt{5}}{2} \text{ and } \lambda_2 = \frac{1 - \sqrt{5}}{2}$$

Consider the following C++ methods for displaying the n th Fibonacci number.

```
// Method 1
int fibonacci(int n) {
    if (n<=1) return n;
    else return fibonacci(n-1) + fibonacci(n-2);
}
cout << fibonacci(n) << endl; // call to function in main program

// Method 2
float l_1 = (1.0 + sqrt(5.0))/2.0, l_2 = (1.0 - sqrt(5.0))/2.0;
cout << (pow(l_1, n) - pow(l_2, n))/sqrt(5.0) << endl;
```

For each method:

- (a) State its algorithmic complexity. [5]
- (b) Comment on its numerical accuracy. [5]

END OF PAPER

Part IA 2015

Paper 4: Mathematical Methods

Numerical Answers

Section A:

1 $x^n = \frac{4^{n+1}}{3} - \frac{1}{3}$

2 (a) -1 (b) -7/6

3 Eigen values are 9, 5; Eigen vectors are (3,1) and (-1, 1)

4 (c) $f = 5000/2\pi$ Hz

5 (c) 9.526 ± 9.47

Section B:

9 (5, 0) – Minimum; (-5, 0) – Maximum; (3, 4) & (-3, -4) – Saddle points

Section C:

11 (a) 267348 bytes

12 (a) Method 1 is $O(2^n)$ (upper bound), Method 2 is $O(1)$